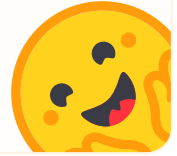


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bitsandbytes

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bitsandbytes enables accessible large language models via k-bit quantization for PyTorch.

bitsandbytes provides three main features for dramatically reducing memory consumption for inference and training:

- 8-bit optimizers uses block-wise quantization to maintain 32-bit performance at a small fraction of the memory cost.
- `LLM.int8()` or 8-bit quantization enables large language model inference with only half the required memory and without any performance degradation. This method is based on vector-wise quantization to quantize most features to 8-bits and separately treating outliers with 16-bit matrix multiplication.
- QLoRA or 4-bit quantization enables large language model training with several memory-saving techniques that don't compromise performance. This method quantizes a model to 4-bits and inserts a small set of trainable low-rank adaptation (LoRA) weights to allow training.

License

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Installation →